

All Your Clicks Belong to Me: Investigating Click Interception on the Web

All Your Clicks Belong to Me: Investigating Click Interception on the Web - Author not stated, [usenix.org](https://www.usenix.org).

- Published: date not stated
- Original: <<https://www.usenix.org/conference/usenixsecurity19/presentation/zhang>>
- Preserved from: <https://www.usenix.org/conference/usenixsecurity19/presentation/zhang> (live) on 2026-08-08
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All Your Clicks Belong to Me: Investigating Click Interception on the Web

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Click is the prominent way that users interact with web applications. For example, we click hyperlinks to navigate among different pages on the Web, click form submission buttons to send data to websites, and click player controls to tune video playback. Clicks are also critical in online advertising, which fuels the revenue of billions of websites. Because of the critical role of clicks in the Web ecosystem, attackers aim to intercept genuine user clicks to either send malicious commands to another application on behalf of the user or fabricate realistic ad click traffic. However, existing studies mainly consider one type of click interceptions in the cross-origin settings via iframes, i.e., clickjacking. This does not comprehensively represent various types of click interceptions that can be launched by malicious third-party JavaScript code.

In this paper, we therefore systematically investigate the click interception practices on the Web. We developed a browser-based analysis framework, Observer, to collect and analyze click related behaviors. Using Observer, we identified three different techniques to intercept user clicks on the Alexa top 250K websites, and detected 437 third-party scripts that intercepted user clicks on 613 websites, which in total receive around 43 million visits on a daily basis.

We revealed that some websites collude with third-party scripts to hijack user clicks for monetization. In particular, our analysis demonstrated that more than 36% of the 3,251 unique click interception URLs were related to online advertising, which is the primary monetization approach on the Web. Further, we discovered that users can be exposed to malicious contents such as scamware through click interceptions. Our research demonstrated that click interception has become an emerging threat to web users.

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@inproceedings {235477, author = {Mingxue Zhang and Wei Meng and Sangho Lee and Byoungyoung Lee and Xinyu Xing}, title = {All Your Clicks Belong to Me: Investigating Click Interception on the Web}, booktitle = {28th USENIX Security Symposium (USENIX Security 19)}, year = {2019}, isbn = {978-1-939133-06-9}, address = {Santa Clara, CA}, pages = {941--957}, url = {https://www.usenix.org/conference/usenixsecurity19/presentation/zhang}, publisher = {USENIX Association}, month = aug }
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